Forecasting Severity-Specific EMS Demand for Ambulance Staffing and Deployment in New York City

1. Executive Summary

The problem: New York City's ambulance system is operating under increasing response-time and capacity pressure. From FY2021 to FY2025, the average end-to-end ambulance response to life-threatening emergencies grew from 9 minutes 34 seconds to 11 minutes 21 seconds, while daily ambulance in-service hours dropped by more than 1,000 and average hospital turnaround rose above 40 minutes.[1] For the most time-critical patients, these minutes carry clinical weight: in witnessed out-of-hospital ventricular fibrillation arrest, every extra minute before the first shock is linked to roughly 6% lower odds of surviving to discharge.[2] FDNY applies call severity effectively once a call arrives, but staffing and unit positioning are still set without a forward-looking estimate of how much demand of each severity to expect, where, and when.

The proposed solution: An AI- and BI-enabled demand outlook that converts Computer Aided Dispatch (CAD) incident history into forecasts of incident counts by severity segment, dispatch area, and time window. A seven-day forecast supports tour staffing; a same-day forecast with threshold alerts supports unit positioning and surge decisions. Both are delivered through a BI dashboard that places each forecast beside historical patterns, its uncertainty range, and its past accuracy. The system runs alongside, not inside, the 911 dispatch path and leaves every operational decision with EMS personnel.

Expected value: The system is designed to place the right type of ambulance capacity in the right area before calls arrive. The largest potential benefit is for Segment 1 emergencies, which make up fewer than 2% of incidents[1] and are therefore easy to overlook in aggregate planning. Expected gains include shorter and more consistent response times, better alignment of paramedic (ALS) resources with high-severity demand, narrower gaps between boroughs, and better use of the $440.3 million EMS appropriation.[1] The system does not add ambulances; it helps FDNY use the ones it has more deliberately.

Recommendation: Approve an 18-month, gated program: two months to secure governance and data access, three months of "shadow mode" in which forecasts are scored but not acted on, a 12-month pilot in selected dispatch areas measured against matched comparison areas, and citywide expansion only if predefined targets are met. Because each phase ends in a go/no-go decision, FDNY commits resources in stages and continues only when the evidence supports it.

2. Healthcare Problem & Organizational Context

The healthcare problem: FDNY responded to 1,615,531 ambulance medical incidents in FY2025.[1] Each 911 medical call is assigned one of eight severity segments. Segment 1 (for example, cardiac arrest or choking) receives both an advanced life support (ALS) and a basic life support (BLS) unit, Segments 2–3 generally receive ALS, and Segments 4–8 receive BLS.[3] Volume is concentrated at the low-acuity end: FDNY handled about 1,173,400 BLS calls and 464,500 ALS calls in FY2024.[3] Segment 1 accounted for only 28,320 incidents in FY2025.[1] The problem this proposal addresses is not how calls are triaged, but the absence of a near-term, severity-specific view of future demand to guide how capacity is staffed and positioned in advance.

Organizational setting: The 911 EMS system is led by FDNY but operated by several parties. Ambulances run by voluntary hospital networks under agreements with FDNY make up about one-third of the units answering 911 calls,[3] and fire companies responded to 371,517 medical incidents in FY2025.[1] Of FDNY's $2.88 billion modified budget for FY2025, $440.3 million was appropriated specifically for EMS.[1] FDNY therefore coordinates a mixed fleet, much of which it does not staff directly, under significant budget and capacity pressure.

Current process: A call is created at 911 intake, classified into a severity segment from the information available at that moment, and dispatched according to priority and required resource type.[3] The incident

then moves through unit assignment, scene arrival, transport, hospital arrival, and closure, and CAD timestamps each step.[4] At the management level, FDNY tracks response times and moves resources toward areas running slower than average.[1] Recent initiatives show an organization willing to test new approaches: turnaround fell by roughly 10.5 minutes at the ten NYC Health + Hospitals sites in the Hospital Liaison Unit pilot, and Paramedic Response Units, which do not transport patients, can see more patients in each tour.[1]

Limitations of the existing approach: Four limitations stand out. First, the process is reactive: severity becomes actionable only once a call is received, and reallocation follows observed performance, so gaps are corrected after they appear. The public FDNY and Independent Budget Office reports reviewed here describe past performance but provide no forward forecast of demand by severity and area; any internal tools are not publicly documented. Second, rare high-severity demand is hidden in aggregate counts, and a citywide volume forecast can look accurate while missing Segment 1 entirely. Third, capacity is tight. Over the past decade, ALS calls grew 31% while paramedic staffing grew 9%,[3] and each ambulance stays unavailable for the full service cycle, including a 40-minute average hospital turnaround.[1] Fourth, performance varies by geography: 68% of BLS calls in Staten Island were reached within 10 minutes in FY2024, compared with 48% in the Bronx.[3]

Why it matters: Research supports the value of closing this gap. Studies in Calgary and Toronto showed that EMS call volumes follow forecastable daily, weekly, and seasonal patterns,[5,6] but focused on total volume rather than severity. Liu et al. optimized next-day and next-week ambulance allocation in New York City while treating future demand as known and identified demand prediction as the missing prior step.[7] This proposal supplies that step, at the severity level where response delays are most costly.

3. AI & BI Solution Design

Solution architecture: The solution has four layers (Figure 1). A *data foundation* ingests and validates CAD incident records, aggregates them into counts, and builds forecasting features. An *AI forecasting layer* compares candidate models and produces severity-specific forecasts with uncertainty ranges. A *BI and delivery layer* presents forecasts, historical context, and alerts. A *decision layer* comprises the people who act on the information and the governance committee that oversees it. A feedback loop returns actual incident counts to the system so that accuracy is monitored and models are retrained on schedule.

Data sources and integration: The system draws mainly on FDNY's EMS Incident Dispatch Data, which the EMS CAD system generates and NYC Open Data publishes for 2005 onward.[4] Each record carries the call type and severity at intake and at closure, timestamps for each operational step, response-time fields with validity flags, and location identifiers such as borough and dispatch area.[4] Because the public feed is refreshed only annually,[4] it is suitable for building and validating models but not for live forecasting; operational use requires a timely, read-only internal CAD extract under a data-sharing agreement. Records are aggregated into hourly counts by severity segment and dispatch area. Where Segment 1 counts are too sparse for stable hourly estimates, forecasts are pooled to wider time windows or to the borough level. Forecasts are based on the initial severity assigned at intake, since that classification determines which unit is sent, and the feature set is restricted to information known before the forecast origin, so that details recorded later in an incident cannot leak into a forecast. The pipeline reads from CAD but never writes to it; FDNY’s call-intake, triage, and dispatch software is untouched.

AI technologies: The forecasting task is predictive analytics on structured time-series data. Three model families are compared on equal terms: a seasonal-naive baseline (the same hour in the previous week), statistical count models that capture recurring hour, day, and seasonal patterns, and gradient-boosted trees (for example, XGBoost), which can model interactions among recent demand, calendar effects, and geography. Features include lagged counts, rolling averages, hour of day, day of week, month, holidays, and area identifiers. Models are validated with rolling-origin evaluation, which always trains on the past and tests on the future, and error is reported separately for each severity segment so that strong performance on common BLS calls cannot mask weak performance on rare Segment 1 calls. Every forecast is issued

with a prediction range, not just a single number. The design deliberately leaves out deep learning, natural language processing, and clinical decision support: the data are tabular, the decisions are operational rather than clinical, and simpler models are easier to explain and audit.

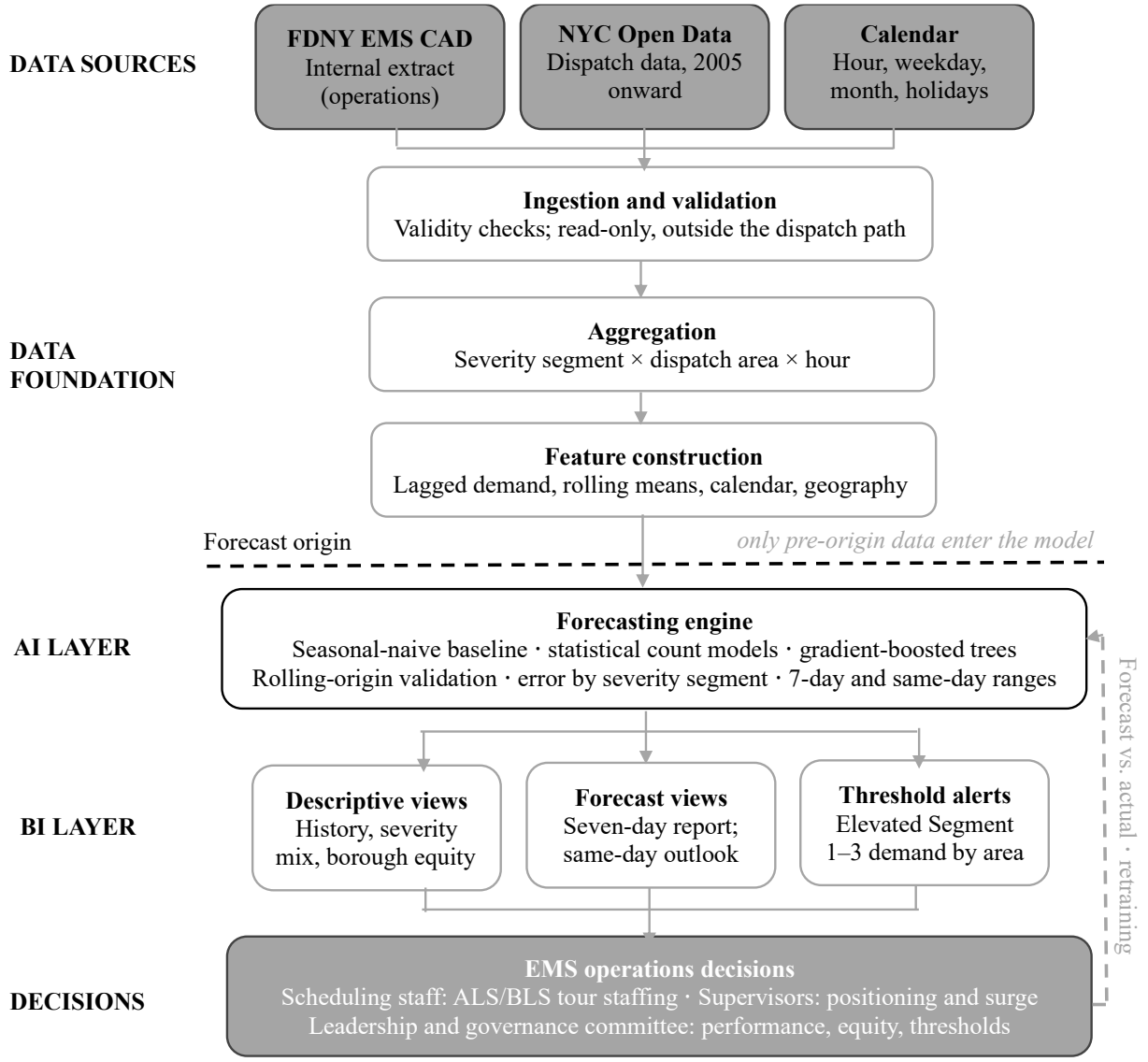


Figure 1. Integrated AI & BI solution architecture. Only data available before the forecast origin (dashed line) enter the model; actual outcomes feed back for monitoring and retraining.

Forecast horizons and outputs: Two forecast products are issued. The *seven-day forecast* is produced weekly and gives expected incident counts for each severity segment and dispatch area in four-hour blocks aligned with tour planning. The *same-day forecast* is produced before each shift and refreshed every few hours at hourly resolution. For each segment and area, the production model is whichever candidate had the lowest rolling-origin error; if no candidate outperforms the seasonal-naive baseline for a given segment in validation, the baseline remains the production method for that segment, preventing unnecessary model complexity when a more advanced method has not demonstrated superior performance. As an initial operating rule, an alert could be raised when forecast Segment 1–3 demand for an area and time window exceeds that area's historical 90th percentile for the same hour and weekday. This threshold is a design

assumption, not an established EMS standard; operations leadership would tune it during shadow mode, balancing missed surges against alert fatigue.

BI technologies: An enterprise BI platform such as Tableau or Power BI provides four views: (1) a descriptive view of historical volume, severity mix, response times, seasonality, and anomalies; (2) a forecast view that shows each area's expected demand beside its seasonal baseline and range; (3) an equity view that tracks response performance by borough; and (4) a model-health view that shows forecast-versus-actual accuracy by segment. Scheduled reports deliver the seven-day outlook, and rule-based alerts flag areas and time windows in which forecast Segment 1–3 demand exceeds thresholds set by operations leadership. Figure 2 shows an illustrative same-day screen.

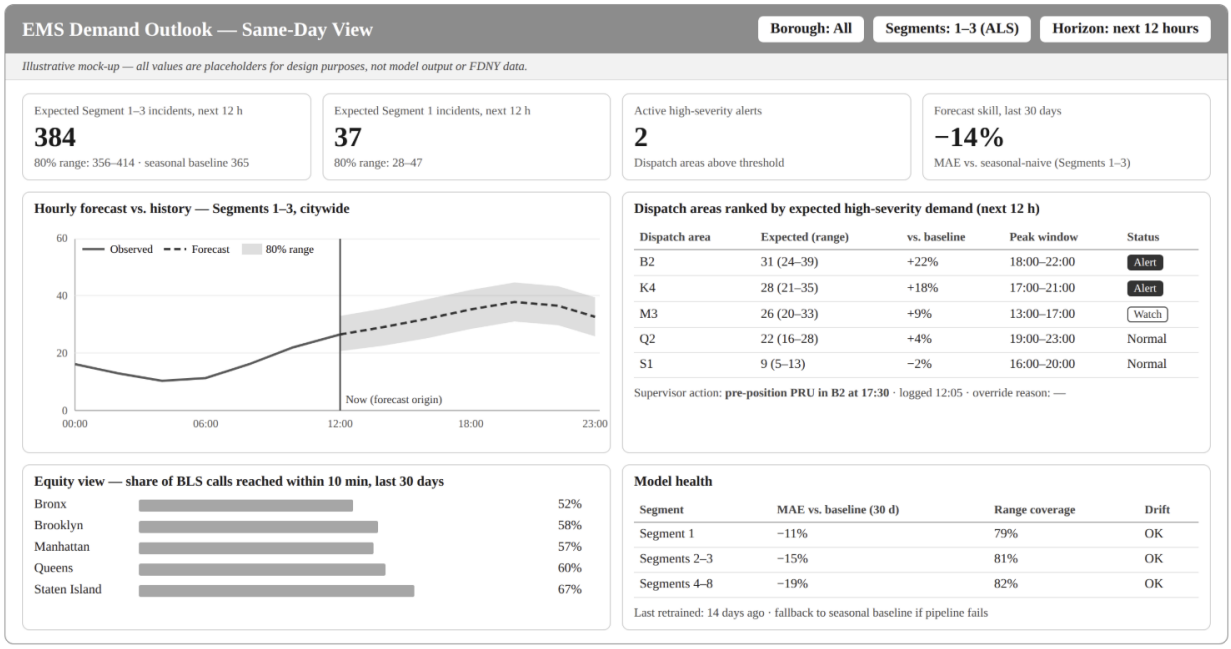


Figure 2. Conceptual dashboard mock-up (same-day view). Values are placeholders for illustration only.

How AI and BI work together: Neither component is sufficient alone. BI without forecasting remains retrospective: it shows where performance slipped yesterday. A forecast without BI is an unexplained number that users have little reason to trust. Together, the AI layer supplies the forward view (what demand is likely, where, and how uncertain the estimate is), while the BI layer supplies context (how that compares with normal patterns), accountability (how accurate past forecasts were), and fairness checks (whether any borough is falling behind). This pairing turns a prediction into a decision that can be reviewed and explained.

Future workflow: The system adds two planning loops ahead of the existing call-handling sequence without changing the sequence itself (Figure 3). In the *weekly staffing loop*, scheduling staff review the seven-day forecast by segment and area and adjust the ALS/BLS mix and geographic coverage of upcoming tours. In the *same-day positioning loop*, dispatch supervisors review the outlook at shift start and receive alerts during the shift. They then decide whether to pre-position units, and Paramedic Response Units suit this role because they do not transport patients.[1] After each cycle, forecast-versus-actual results feed the next one.

User interaction: Consider a typical afternoon. At 12:00, a dispatch supervisor opens the same-day view and sees that expected Segment 1–3 demand in one Bronx dispatch area for 18:00–22:00 is well above its seasonal baseline and has triggered an alert. After checking unit availability, the supervisor pre-positions a Paramedic Response Unit nearby at 17:30 and records the decision in one click; if they choose not to act, they record a reason instead. Scheduling staff interact weekly through the seven-day report, and EMS

leadership reviews monthly performance and equity summaries. Field crews use no new interface and experience the system only through supervisors' positioning decisions.

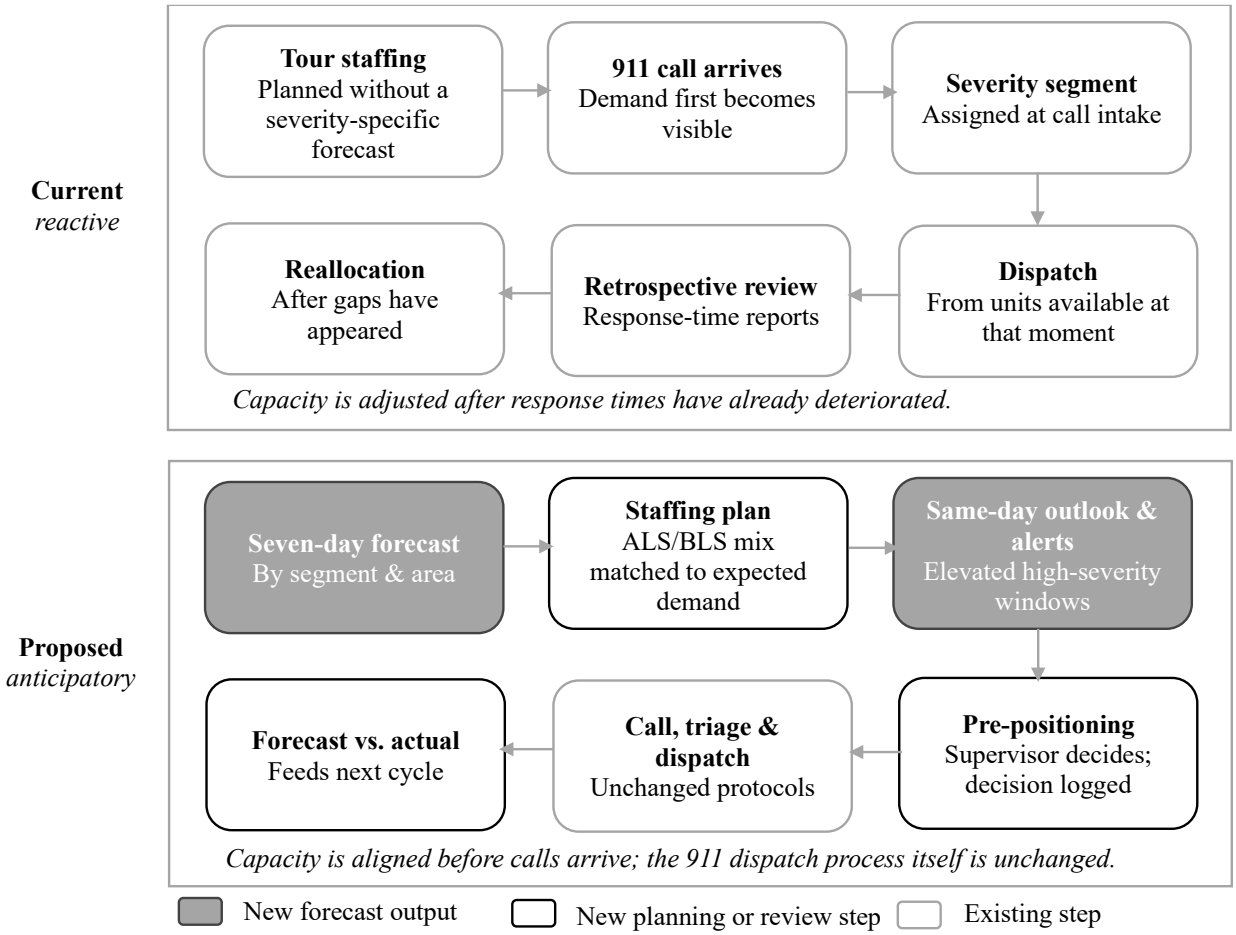


Figure 3. Before-versus-after workflow. Forecasts inform staffing and positioning before calls arrive; 911 dispatch protocols are unchanged.

4. Implementation Readiness & Governance

Organizational readiness: Readiness is moderate to high. FDNY already manages EMS through published performance metrics and geographic reallocation,[1] so forecasting extends an existing habit rather than introducing a new one. The Hospital Liaison Unit shows a pilot-and-measure culture that suits a phased rollout.[1] Technical readiness is moderate, held back mainly by three gaps: timely access to internal CAD data, the lack of hourly unit-availability data (which limits the system to forecasting demand rather than the full supply-demand balance), and a capacity constraint that forecasting cannot remove.

Capacity remedies: Because forecasting can redistribute capacity but not create it, the pilot will test forecasts alongside four candidate operational levers available to FDNY: (1) forecast-informed placement of Paramedic Response Units, which handle more patients per tour since they do not transport;[1] (2) prioritizing Hospital Liaison Unit coverage at hospitals serving areas of high forecast demand, since shorter turnaround returns ambulance hours to service;[1] (3) routing eligible low-acuity patients to the 911 telemedicine pathway during forecast peaks;[1] and (4) aligning overtime and staggered tour start times with forecast peak windows. Each lever's contribution will be evaluated during the pilot, and where demand persistently exceeds what they can absorb, that evidence can support requests for additional paramedic positions.

Governance and regulatory considerations: Governance follows the four functions of the NIST AI Risk Management Framework.[8] *Govern:* an executive sponsor in EMS Operations owns the program, supported by a quarterly oversight committee that sets alert thresholds and draws members from operations, dispatch, the medical director's office, analytics and IT, labor, and the partner hospital fleets. *Map:* known risks, such as historical calls capturing who dialed 911 rather than who truly needed care, are documented before deployment. *Measure:* accuracy is tracked monthly by segment and borough. *Manage:* models are retrained quarterly, and sooner if a segment's four-week error rises more than 20% above its shadow-mode level or the model fails to beat the seasonal baseline for two consecutive months; these initial tolerances are set and adjusted by the governance committee. The dashboard falls back to the seasonal baseline if the pipeline fails. For transparency, each forecast is displayed with its seasonal baseline and prediction range, the gradient-boosted model reports which inputs most influence each prediction, and model documentation records data, horizons, validation results, and known limitations. Because the tool could influence public-safety resource allocation, it should be reviewed for reporting under New York City's algorithmic-tool disclosure requirements (Local Law 35 of 2022).[9]

Privacy and security: Models use aggregated counts rather than patient records, and the public dataset already removes incident locations and coarsens geography.[4] Internal CAD extracts contain incident-level detail and therefore require role-based access, encryption in transit and at rest, short retention periods, and review for applicable HIPAA, City, and FDNY privacy requirements before use. Small Segment 1 counts are suppressed in any output shared outside FDNY.

Resource requirements: The program can build largely on existing data and BI infrastructure but requires dedicated analytical, governance, and operational support. An executive sponsor and a product owner within EMS Operations provide direction. A small analytics team of a data engineer, a data scientist, and a BI developer builds and maintains the pipeline, models, and dashboards, supported by IT security for access control. The system uses existing CAD data and can run on FDNY's or the City's existing BI platform and standard computing infrastructure, so there is no need to procure a specialized clinical system. Staff time for training and weekly forecast review is the main operational cost, which is why training is role-based and the added workflow step is kept to a single review-and-record action.

Stakeholder adoption and change management: Adoption depends on trust, which is built before anyone must act on a forecast. During shadow mode, forecast-versus-actual results, including misses, are shown openly, dashboards are co-designed with scheduling staff and supervisors, and alert thresholds are tuned to avoid alert fatigue. Role-based training before each phase covers how to read a range and when an override is appropriate. Each borough has a champion as a first point of contact, and users can report any forecast that conflicted with what they saw in the field. Overrides are logged but never penalized, and forecasts are used for deployment, not to evaluate individual crews, which addresses a likely concern for EMTs, paramedics, and their unions. Partner hospital services receive area-level forecasts and a role in pilot planning. The rollout follows the four gated phases in Figure 4.

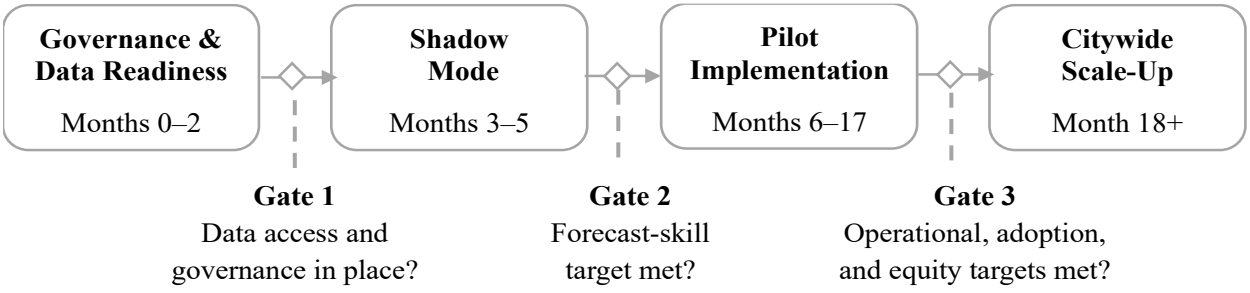


Figure 4. Phased implementation roadmap. Each phase ends at a go/no-go decision gate.

Major risks and mitigation strategies: The principal risks and planned responses are summarized below.

Table 1. Implementation risks and mitigation strategies

Risk	Mitigation
Delayed access to timely internal CAD data	Formal data-sharing agreement required at Gate 1; pipeline and descriptive BI built on public data in the meantime
Unstable forecasts for sparse Segment 1 demand	Pool to wider time windows or borough level; report error by segment; always compare against the seasonal-naive baseline
Sudden demand shifts (e.g., COVID-19[7])	Monthly drift monitoring; retraining if a segment's four-week error rises >20% above its shadow-mode level or fails to beat the baseline for two consecutive months; automatic fallback to baseline
Alert fatigue or low trust among users	Thresholds tuned in shadow mode; misses shown openly; overrides respected and logged, not penalized
Inequity from allocating by predicted volume alone	Minimum coverage floor in every area; accuracy and response performance reported by borough
Labor concern that forecasts drive cuts or crew evaluation	Labor seat on governance committee; forecasts used for deployment only
Expectations exceed what repositioning can deliver	Communicate that forecasts reposition but do not add units; use results to inform capacity requests

5. Organizational Value & Performance Evaluation

Clinical outcomes: By anticipating where high-severity demand will concentrate, the system aims to shorten the interval between a Segment 1–3 call and the arrival of an ALS-capable unit. For cardiac arrest and similar emergencies, where each minute is associated with lower survival,[2] even modest reductions in response time matter. Clinical benefit is expected rather than assumed; it is tested through the response-time KPIs below.

Operational efficiency: Reallocation moves from after-the-fact correction toward advance planning. Scheduling staff match the ALS/BLS mix to expected demand, and supervisors position units before surges rather than during them. Because the system sits outside the dispatch path, it poses no risk to 911 call handling.

Patient experience: Patients may benefit from shorter and more consistent waits if forecast-informed positioning improves response performance. The equity view and the geographic-variation KPI keep attention on neighborhoods with historically slower service.

Organizational performance: Forecasts state expected demand explicitly, allow comparison across areas, and can be checked against actual outcomes. This strengthens decision quality and gives leadership objective evidence for staffing and budget discussions, including where added capacity, not better positioning, is the real need.

Value by stakeholder: Patients stand to gain faster, more consistent access to the right level of care. EMTs and paramedics may benefit from better-planned tours and fewer unplanned surges, easing workload pressure in a workforce that has grown more slowly than demand.[3] Dispatch supervisors gain an early warning rather than discovering surges as they happen. Leadership gains an auditable link between planning decisions and outcomes. Partner hospital ambulance services gain visibility into expected demand in the areas they cover.

Evaluation design and KPIs: KPIs cover three levels: whether forecasts are accurate, whether they are used, and whether outcomes improve. Pilot areas are compared with matched comparison areas over the same period, so that citywide trends are not mistaken for program effects. Numeric targets are management-defined pilot criteria that the governance committee confirms after shadow mode and reviews annually.

Guarding against unintended effects: Because repositioning a unit into one area can leave a neighboring area thinner, response times in areas adjacent to the pilot are monitored alongside the pilot and comparison areas. Override logs are reviewed quarterly to learn whether users disagree with forecasts in systematic ways, which may reveal a model weakness or a local factor the model does not capture. The equity KPI requires that no area's performance declines, so improvement in the pilot as a whole cannot be achieved at the expense of a single neighborhood.

Table 2. Performance evaluation KPIs

KPI	Level	Target	Baseline	Timeframe
Segment-level forecast skill (MAE)	Model	≥10% lower MAE than seasonal-naive in every segment	Seasonal-naive MAE (shadow mode)	End of shadow mode, then monthly
Forecast use in planning	Adoption	Forecast reviewed and decision logged in ≥80% of pilot cycles and shifts	0% (new)	Pilot month 6
End-to-end ambulance response time to life-threatening emergencies	Outcome	≥30 seconds lower in pilot vs. comparison areas	11:21 citywide, FY2025[1]	12 months of pilot
ALS-capable unit response share	Outcome	≥3 percentage points above pilot-area baseline, net of change in comparison areas	Measured in shadow mode	12 months of pilot
Geographic response-time variation	Equity	Gap narrowed by ≥5 percentage points; no area declines	Pilot-area gap (boroughs 48%–68%, FY2024[3])	12 months of pilot

6. Strategic Recommendations

Overall recommendation: FDNY should adopt the severity-specific demand forecasting and BI system through the gated program described above, approving Phases 1 and 2 now and committing to the pilot only if the forecast-skill target is met at Gate 2.

Why adopt: The solution addresses a clear, documented gap (the lack of a forward view of severity-specific demand) at a time of rising response times and constrained capacity.[1,3] It builds on practices FDNY already uses, requires no change to dispatch protocols, keeps humans in control, and is evaluated against transparent, predefined targets. The gated design limits downside: if forecasts are not accurate enough, the program stops before operations are affected.

Short-term benefits (0–12 months): A validated data pipeline and descriptive dashboards give leadership a clearer, segment-level view of demand, response performance, and geographic variation; this value holds even if the pilot is not approved. Shadow mode produces objective evidence of forecast accuracy and establishes KPI baselines.

Long-term benefits (12+ months): Anticipatory staffing and positioning become routine, ALS resources align more closely with high-severity demand, borough gaps narrow, and leadership gains an auditable basis for resource and budget decisions.

Scalability and future enhancements: The architecture scales by adding dispatch areas and boroughs without redesign. Future enhancements include (1) adding external predictors such as weather, heat alerts, and major public events; (2) integrating unit-availability data to forecast the gap between demand and supply rather than demand alone; (3) connecting forecasts to allocation-optimization models such as those developed by Liu et al.[7]; (4) forecasting hospital turnaround alongside the Hospital Liaison Unit program;

and (5) sharing area-level forecasts with partner hospital ambulance services. Because the approach relies on standard CAD data, it could also be adapted by other large urban EMS systems.

Key assumptions and limitations: The proposal forecasts demand, not the full balance of supply and demand, because the public data do not record hourly unit availability. Forecast skill for Segment 1 is the least certain element and may require coarser time or geographic resolution than other segments. Historical 911 records reflect how people used the system, not their underlying need. Most importantly, forecasting helps position existing capacity; it cannot offset a structural shortfall in paramedic staffing. These limitations are why the pilot is measured against comparison areas and why continued investment depends on the evidence.

Decisions requested of leadership: To begin Phase 1, leadership is asked to (1) designate an executive sponsor and product owner in EMS Operations; (2) authorize a data-sharing agreement for a timely, read-only CAD extract; (3) resource the analytics team for Phases 1 and 2; (4) charter the governance committee, including labor and partner hospital representation; and (5) direct the committee to recommend pilot and comparison areas before Gate 2.

Closing: FDNY already responds to severity once a call arrives. This proposal gives it the ability to prepare for severity before the call comes in, extending severity-based decision-making upstream to the patients for whom minutes matter most.

7. References

1. New York City Mayor's Office of Operations. *Mayor's Management Report, Fiscal 2025: Fire Department.* City of New York; 2025. Accessed August 28, 2026. https://www.nyc.gov/assets/operations/downloads/pdf/mmr2025/fdny.pdf
2. Stieglis R, Verkaik BJ, Tan HL, Koster RW, van Schuppen H, van der Werf C. Association between delay to first shock and successful first-shock ventricular fibrillation termination in patients with witnessed out-of-hospital cardiac arrest. *Circulation.* 2025;151(3):235-244. doi:10.1161/CIRCULATIONAHA.124.069834
3. Gudino V, Stuart C, Dolan E. *Trends From a Decade: 2014–2024 Response Times to Medical Emergencies Getting Worse.* New York City Independent Budget Office; March 2026. Accessed August 28, 2026. https://www.ibo.nyc.gov/assets/ibo/downloads/pdf/public-safety/2026/2026-march-trends-from-a-decade.pdf
4. Fire Department of the City of New York. EMS Incident Dispatch Data. NYC Open Data. Updated July 15, 2026. Accessed September 22, 2026. https://data.cityofnewyork.us/Public-Safety/EMS-Incident-Dispatch-Data/76xm-jjuj
5. Channouf N, L'Ecuyer P, Ingolfsson A, Avramidis AN. The application of forecasting techniques to modeling emergency medical system calls in Calgary, Alberta. *Health Care Manag Sci.* 2007;10(1):25-45. doi:10.1007/s10729-006-9006-3
6. Matteson DS, McLean MW, Woodard DB, Henderson SG. Forecasting emergency medical service call arrival rates. *Ann Appl Stat.* 2011;5(2B):1379-1406. doi:10.1214/10-AOAS442
7. Liu J, Ouyang R, Chou C-A, Griffin J. An analytical approach for dispatch operations of emergency medical services: a case study of COVID-19. Oper Res Forum. 2023;4(2):44. doi:10.1007/s43069-023-00218-3
8. National Institute of Standards and Technology. *Artificial Intelligence Risk Management Framework (AI RMF 1.0).* NIST AI 100-1. National Institute of Standards and Technology; 2023. doi:10.6028/NIST.AI.100-1
9. New York City Office of Technology and Innovation. *Agency Compliance Reporting of Algorithmic Tools: Calendar Year 2024.* City of New York; 2025. Accessed September 22, 2026. https://www.nyc.gov/assets/oti/downloads/pdf/reports/2024-algorithmic-tools-report.pdf